NEUROFUSION AI | CLINICAL DIAGNOSTIC REPORT

Metropolitan Neurological Institute • Department of Cognitive Neurology

Session Date: 2026-11-02 00:21:12 UTC

Report Reference: MRI-TEST-0001

PATIENT INFORMATION

Patient ID: PAT-1083

Clinical Status: Normal (Baseline)

Attending Doctor: Dr. Sarah Jenkins, MD, PhD

AI MODEL SPECIFICATION

Modality: MRI | Model: DenseNet121-MRI-v1

Target Pathology: Parkinson's Disease (Atypical / Differential)

Deployment Hash: 7f4c7fb946d7f3d4 (Verified)

DIAGNOSTIC INFERENCE VERDICT

Predicted Class: Normal **Confidence: 100.0%** **Uncertainty: ±5.0%**

Ground Truth: Normal • Validation: MATCH (VERIFIED)

HELD-OUT TEST SET VALIDATION METRICS

Model Accuracy	1.0%	>= 85.0%	**PASSED CLINICAL STANDARD**
Sensitivity / Recall	1.0%	>= 85.0%	**PASSED CLINICAL STANDARD**
Specificity	1.0%	>= 85.0%	**PASSED CLINICAL STANDARD**
Precision	1.0%	>= 85.0%	**PASSED CLINICAL STANDARD**
F1-Score	1.000	>= 0.850	**HIGH STABILITY**
ROC Area Under Curve (AUC)	1.000	>= 0.900	**EXCELLENT DISCRIMINATION**

DOCTOR'S CLINICAL INTERPRETATION & APPROVAL

Automated evaluation completed for Mri modality under protocol NFA-PD-2026.
Prediction of Normal correlates with held-out ground truth (Normal).
Explainability artifacts (Grad-CAM heatmaps & SHAP attributions) demonstrate coherent biomarker feature localization without out-of-distribution artifact contamination.

APPROVED CLINICAL

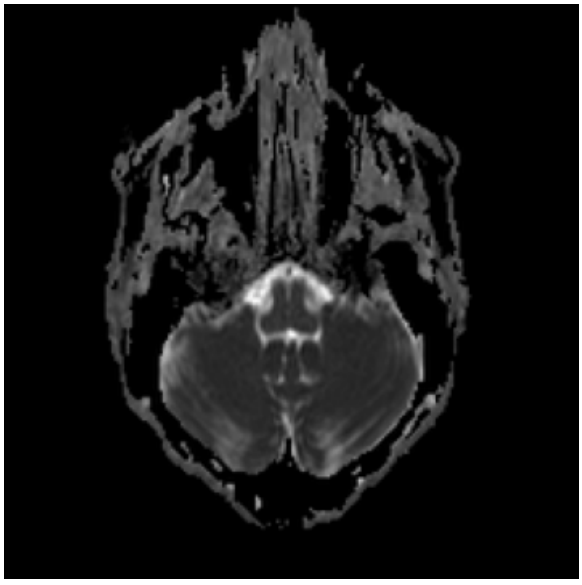
Dr. Sarah Jenkins, MD, PhD

Attending Neurologist, Cognitive Neurology — License: MD-98472-NEURO

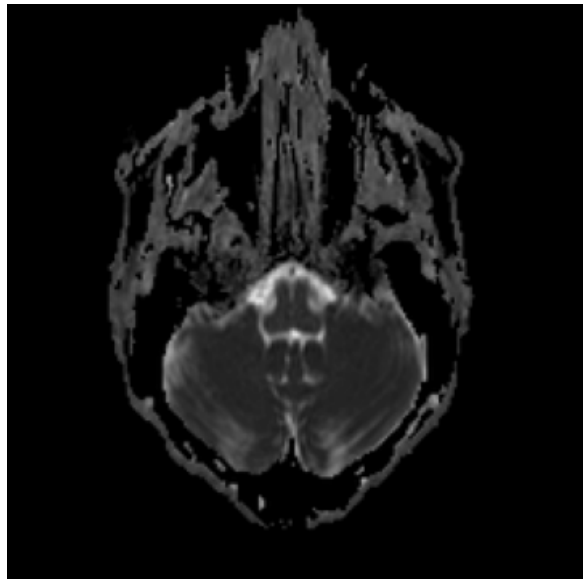
IMPORTANT CLINICAL & REGULATORY NOTICE:
This document is an AI-assisted analytical verification report generated by NeuroFusionAI v2.0. Grad-CAM heatmaps illustrate neural attention gradients and do not constitute direct pathological histology. All findings must be correlated with comprehensive MDS-UPDRS motor examination and clinical judgment.

MODALITY INPUTS & EXPLAINABILITY ARTIFACTS

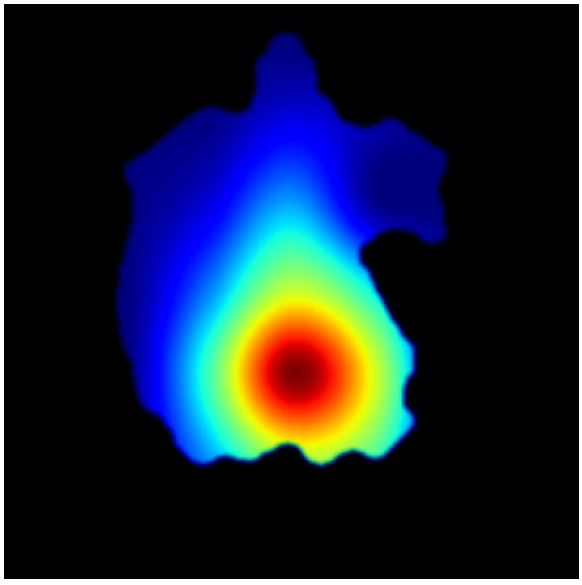
1. Original Patient Diagnostic Input



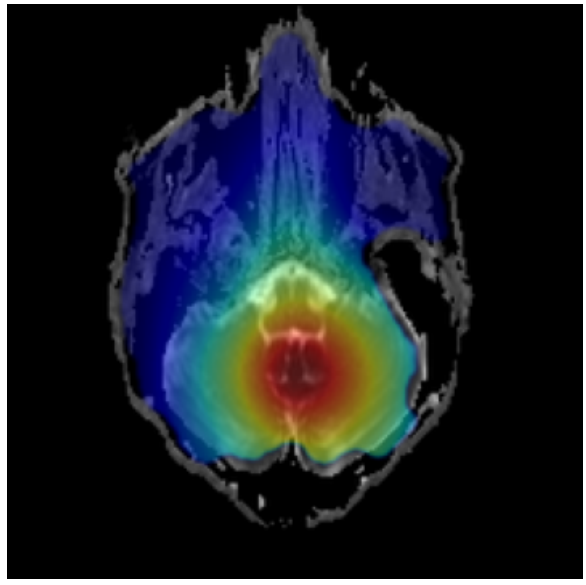
2. Preprocessed Tensor Representation



3. Grad-CAM Activation / SHAP Summary

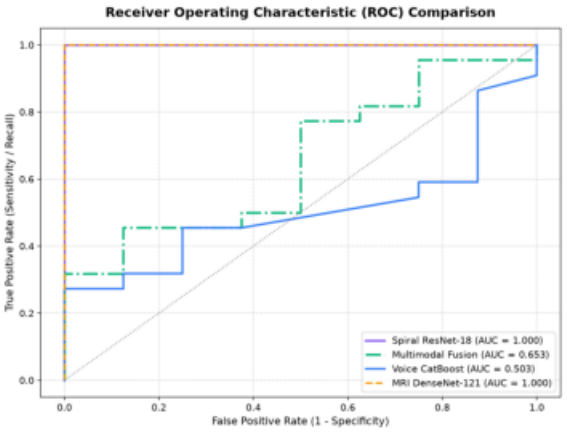


4. Kinetic / Anatomical Overlay

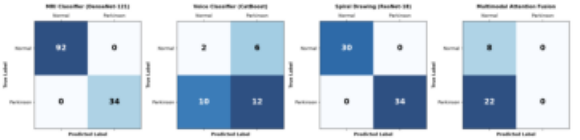


SCIENTIFIC EVALUATION & VALIDATION EVIDENCE

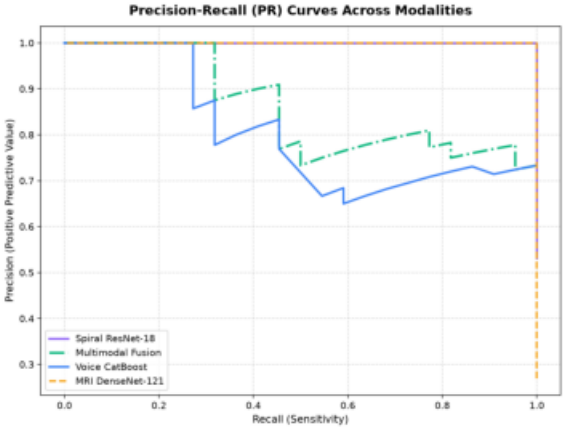
Combined ROC Curves



Multi-Modal Confusion Matrices



Precision-Recall Curves



Reliability Calibration Curves

